



Applied Machine Learning

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Abstract

This report presents a machine learning case study that leverages a Random Forest classifier to predict the severity level of gun violence incidents in the United States. Severity has been defined under three categories: low (class 0), medium (class 1), and high (class 2). The dataset used for this project contains information about gun violence incidents, including the number of killed or injured victims, the number of guns involved, as well as categorical attributes such as state, whether the gun was stolen, and gun type categories (Pistol, Shotgun, Rifle, etc.). Through systematic data cleaning, feature engineering, and hyperparameter tuning, the Random Forest model achieved an overall accuracy of approximately 63.4%. In particular, the model displayed balanced performance for classes 0 (low severity) and 1 (medium severity), but encountered difficulties predicting class 2 (high severity). A critical discussion on model performance, ethical considerations, class imbalance, and possible improvements is also included. Finally, the report compares the Random Forest model's performance with results from three alternative classification methods: Naive Bayes, Logistic Regression, and XGBoost.

1. Introduction

Gun violence is a pervasive issue in many parts of the world, particularly in the United States. Understanding factors that contribute to varying degrees of gun violence severity can provide valuable insights to policymakers, emergency responders, and community organizations aiming to mitigate or prevent further incidents. Traditional statistical methods have been used to explore these patterns. However, the availability of large datasets and advancement in computing power have paved the way for machine learning algorithms to enhance predictive analytics in this domain.

This case study specifically addresses the problem of classifying the severity of gun violence into low, medium, and high levels. Building a robust predictive model can help authorities prioritize resources, inform targeted prevention programs, and facilitate swifter emergency responses based on projected severity. This report uses Random Forest—a powerful ensemble machine learning technique—as the primary classification model. Random Forest often outperforms simpler models such as single decision trees due to its bagging of multiple trees, inherent feature importance estimation, and robustness to overfitting.

The remainder of this report is structured as follows: Section 2 covers the dataset, data cleaning techniques, and feature selection. Section 3 discusses the experimental setup, including the creation of training and testing subsets. Section 4 summarizes results and model performance metrics. Section 5 compares the Random Forest model with other established methods (Naive Bayes, Logistic Regression, and XGBoost), highlighting the strengths and limitations of each. Section 6 presents a discussion, ethical considerations, conclusions, and potential avenues for future work.

2. Data Exploration and Feature Selection

2.1 Dataset Description

The dataset used in this study captures gun violence incidents across the United States. Each incident comprises a range of features:

1. **State:** The U.S. state where the incident occurred.
2. **n_killed:** Number of people killed in the incident.
3. **n_injured:** Number of people injured.
4. **gun_stolen:** Status indicating if the gun was stolen, unknown, or not stolen.
5. **gun_type:** Type(s) of firearms involved (e.g., Pistol, Shotgun, Rifle, Other).
6. **n_guns_involved:** Total number of firearms associated with the incident.

Several columns containing irrelevant information (exact street address, incident URLs, participant relationships, etc.) were removed to maintain focus on key variables. The severity level was eventually derived from the combined total of killed and injured victims (i.e., total casualties).

2.2 Data Cleaning and Preprocessing

Initial data cleaning steps included:

- **Date Conversion:** Converting the date column from string format to a Date object.
- **Missing Values:** Treating empty strings as NA and subsequently dropping rows missing critical columns (e.g., gun_stolen, n_guns_involved). This was crucial to ensure that the training data contained complete information for the features deemed essential.
- **Removing Irrelevant Columns:** Columns unlikely to influence severity prediction (e.g., URL links, location descriptions) were excluded, reducing dimensionality.

After these steps, a subset of data was selected containing only relevant features for model training.

2.3 Feature Engineering

Feature engineering was conducted to create meaningful inputs for the Random Forest model:

1. **Gun Type Categorization**
Each incident could involve multiple firearms. The original gun_type field listed types in a concatenated string. Using custom mappings, gun types were categorized into “Unknown_type,” “Pistol,” “Shotgun,” “Rifle,” and “Other.” Binary indicator columns (one-

hot encoding) were then created to represent whether a particular gun type was involved.

2. Gun Stolen Status

Similar to gun type, gun_stolen sometimes contained multiple statuses if more than one gun was involved. These were split and converted to binary indicators for “Unknown,” “Stolen,” and “Not-stolen.”

3. Severity Definition

A new variable, n_casualties, was defined as the sum of n_killed and n_injured. This numeric value served as the basis for grouping incidents into severity classes:

- **Class 0 (Low severity):** 0 total casualties
- **Class 1 (Medium severity):** 1 total casualty
- **Class 2 (High severity):** 2 or more total casualties

While alternative binning strategies (e.g., 0–2, 3–5, >5 casualties) were considered, the chosen thresholds best balanced distribution and interpretability in the dataset. Finally, to avoid data leakage, n_casualties was dropped after creating the severity label, ensuring that the algorithm only learns from the categorical severity label rather than the numeric sum directly.

2.4 Exploratory Data Analysis (EDA)

Basic descriptive statistics revealed that:

- Most incidents resulted in zero or one casualty (class 0 or 1).
- Class 2 (high severity) was relatively rare, highlighting a class imbalance.
- Handguns (Pistol) were the most common gun type, while high-capacity rifles were less frequent.
- The majority of rows had “Unknown” for gun_stolen, indicating incomplete data in that field.

The final dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve class distribution. A further subset of 20,000 training samples and 4,000 test samples was used to speed up initial experimentation, ensuring the ability to refine the model in a timely manner.

3. Experiments with the Random Forest Model

3.1 Algorithm Selection: Why Random Forest?

Random Forest is an ensemble learning method that integrates multiple decision trees through a bagging strategy. It is recognized for:

- **Robustness:** By averaging predictions over many trees, outlier decisions in any single tree are mitigated.
- **Feature Importance:** Random Forest naturally produces a measure of feature importance, guiding interpretations of model predictions.
- **Handling of Nonlinear Interactions:** Decision trees (and by extension Random Forest) inherently model nonlinear relationships without additional feature transformations.

Given these strengths, Random Forest was an appropriate choice to handle a moderately sized dataset with mixed numeric and categorical predictors.

3.2 Model Training

Training was conducted with the caret package in R:

1. **Cross-Validation:** A 3-fold cross-validation scheme was used to estimate generalization error. Though 5- or 10-fold cross-validation is more common, a smaller number of folds was used to reduce computational overhead.
2. **Hyperparameter Tuning:**
 - **mtry:** This parameter specifies the number of features considered at each split. A grid search was performed with two candidate values (2 and 59).
 - **ntree:** Internally, caret sets a default number of trees for the randomForest package (usually 500), though the optimal number can be further tuned if needed.

3.3 Performance Metrics

The primary metric was **Accuracy**, representing the proportion of correctly classified samples. However, given the class imbalance, additional statistics—**Sensitivity**, **Specificity**, **Precision**, **F1-score**, and **Kappa**—offer deeper insight into how each class is recognized.

4. Results

4.1 Accuracy and Best Hyperparameters

Through cross-validation, the best value of mtry was found to be 59 among the tested options. The accuracy of the model on the test set was **63.4%**,

4.2 Class-Wise Performance

A confusion matrix was produced to evaluate model performance per severity class:

- **Class 0 (Low severity):**
 - **Sensitivity:** ~60.5% – The model identified around 60% of actual class 0 samples correctly.
 - **Specificity:** ~79.4% – Roughly 79% of the non-class 0 samples were correctly identified as belonging to other classes.
- **Class 1 (Medium severity):**
 - **Sensitivity:** ~79.8% – The model displayed relatively strong recall for medium-severity incidents.
 - **Specificity:** ~54.1% – More than half of non-class 1 incidents were correctly identified, but there was room for improvement.
- **Class 2 (High severity):**
 - **Sensitivity:** ~0.5% – This extremely low value indicates the model struggled to correctly identify high-severity cases, which are rare in the dataset.
 - **Specificity:** ~99.7% – The model almost never misclassifies lower-severity incidents as high severity, leading to near-perfect specificity but sacrificing recall significantly.

Although the Kappa statistic of ~0.333 indicates a fair agreement above chance, it also underscores that the skew in class distributions (particularly for class 2) hampers the ability to capture rare but critical high-severity incidents.

4.3 Feature Importance

An examination of feature importance revealed:

1. **n_guns_involved** – The total count of firearms was influential in predicting whether more casualties occurred.
2. **State** – Geographical location offered moderate predictive power, potentially capturing socio-demographic patterns.
3. **Pistol** – The presence of a handgun correlates significantly with incident severity.

4. **Unknown_type, Rifle, Shotgun** – These categories offered additional but lesser predictive utility.
5. **Gun_stolen** (binary flags) – Generally lower importance, possibly due to high “Unknown” rates.

5. Comparison with Other Models

5.1 Naive Bayes

A Naive Bayes model was also trained for the same classification task. It achieved an overall accuracy of **53.99%**, significantly lower than Random Forest’s 63.4%. Key observations:

- **Class 1:** This was predicted with high sensitivity (~94.33%), likely because class 1 forms a large proportion of the dataset, making it a frequent default prediction for Naive Bayes.
- **Class 0 and Class 2:** Naive Bayes struggled, yielding sensitivity values of ~24.63% and ~0.42%, respectively.
- **Class 2:** The model almost never caught high-severity incidents correctly.

While Naive Bayes is known for its computational efficiency and interpretability, it showed limited success for this imbalanced dataset, especially for classes 0 and 2. Its performance is hindered by the strong assumptions of feature independence and the presence of skewed class distributions.

5.2 Logistic Regression

Logistic Regression yielded an overall accuracy of **62.49%**, just below the Random Forest’s 63.4%. The model excelled for:

- **Class 0 (Low severity):** High precision and specificity, with an ROC-AUC of 0.75.
- **Class 1 (Medium severity):** Balanced results with an ROC-AUC of 0.71.

However, it failed to predict Class 2 (high severity) effectively because the dataset’s imbalance caused the model to rarely classify an incident as high severity, resulting in a near-zero true positive rate for that class. Although the linear boundary assumption can work reasonably well, the inherent complexity and imbalance in the dataset hamper logistic regression’s capacity to separate class 2 effectively. Despite its shortfall for the rare class, logistic regression provided a solid baseline with relatively straightforward interpretability and moderate success for the majority classes.

5.3 XGBoost

XGBoost is a boosted ensemble algorithm that typically excels in structured data. Its performance in this study was reported in two versions: **unweighted** and **weighted**.

1. Unweighted Version

- **Accuracy:** ~62.08%
- **Class 0:** Sensitivity of 54.32%; specificity of 82.76%.
- **Class 1:** Sensitivity of 83.53%; specificity of 47.94%.
- **Class 2:** Very poor sensitivity (~0.03%), but perfect specificity.

2. Weighted Version

- **Accuracy:** ~61.07% (slightly lower overall, as weighting tries to correct the imbalance).
- **Class 0:** Sensitivity at 51.79%; specificity at 84.12%.
- **Class 1:** Sensitivity at 83.07%; specificity at 47.71%.
- **Class 2:** Improved from ~0.03% to ~3.88% sensitivity, though still critically low.

While XGBoost demonstrated strong performance for classes 0 and 1, it struggled to capture class 2 due to the pronounced imbalance. Even with reweighting, capturing high-severity incidents remains challenging, though the sensitivity did improve.

5.4 Summary of Comparisons

The Random Forest approach slightly surpasses logistic regression and both variants of XGBoost in terms of raw accuracy. It also offers improved interpretability over XGBoost regarding feature importance. However, none of the models effectively handled class 2, the high-severity category, due to its low frequency in the dataset. Naive Bayes produced the lowest overall accuracy and heavily struggled with classes 0 and 2. Logistic Regression and XGBoost performed moderately, with logistic regression giving a simpler, more interpretable baseline. Meanwhile, XGBoost, particularly with weighting, made small gains in capturing class 2 but did so at the expense of the other classes.

6. Discussion, Conclusions, and Future Work

6.1 Discussion of Findings

Model Performance:

- Random Forest achieved the highest test accuracy (63.4%) among the compared models.
- None of the models performed well for class 2 (high severity), primarily due to a limited number of such cases in the dataset.

Interpretability and Class Imbalance:

- The confusion matrix underlines that medium-severity incidents are more reliably predicted than low- or high-severity ones across most algorithms.
- Class imbalance significantly reduces the ability to capture critical but rare high-severity events. Traditional metrics like overall accuracy can be misleading in these scenarios.

Feature Importance Insights:

- The prominence of `n_guns_involved` correlates with increased severity—unsurprising, as multiple firearms might lead to more casualties.
- Differences in gun type appear moderately predictive, with pistols occurring frequently in medium-level incidents.
- High-severity incidents might involve multiple firearms or rifles, but these were not captured effectively by the model, possibly because truly high-severity cases remain sparse relative to total incidents.

6.2 Ethical, Professional, and Legal Considerations

In deploying a model to predict the severity of gun violence:

1. Ethical Concerns:

- Misclassification of high-severity incidents could negatively impact emergency responders who might deprioritize or misallocate resources.
- Any biases inherent in the dataset (e.g., geographic or demographic factors) may lead to unjust profiling or resource allocation decisions.

2. Data Privacy:

- Detailed gun violence data can contain sensitive personal information. Using and sharing such datasets necessitates strict adherence to privacy and data protection regulations, including anonymization and secure storage.

3. Legal Liability:

- Predictive models that inform real-world actions, like dispatching law enforcement or medical teams, raise concerns about liability if model predictions turn out to be incorrect. Institutions must establish guidelines to use the model's output as one factor among many in decision-making processes.

6.3 Conclusion

This study sought to predict the severity of gun violence incidents using machine learning classification. Through a well-defined preprocessing pipeline, feature engineering, and random forest training with cross-validation, the model performed better than other tested approaches, achieving around 63.4% accuracy. Despite these improvements, the results emphasize that identifying rare but critical high-severity incidents remains a challenge with the given dataset. Overcoming class imbalance demands additional strategies, such as:

- **Resampling Techniques:** SMOTE (Synthetic Minority Oversampling Technique) or undersampling of majority classes may help.
- **Cost-Sensitive Learning:** Assigning heavier penalties for misclassifications of the minority class.
- **Ensemble Approaches:** Combining multiple base learners or using advanced boosting strategies specifically tailored to skewed data.

6.4 Future Work

Several avenues exist to enhance this severity prediction system:

1. **Data Augmentation**
 - Additional historical records or external data (e.g., local crime rates, socioeconomic indicators) may improve predictive power.
2. **Advanced Resampling**
 - SMOTE or ADASYN could generate synthetic samples for high-severity incidents, thereby boosting minority-class presence.
3. **More Granular Severity Definition**
 - Future research might adopt multi-level severity scales (e.g., more than three classes) or separate models for predicting the number of casualties as a regression task.
4. **Hyperparameter Optimization**

- Conduct a more exhaustive search for mtry, ntree, node size, or other advanced parameters to refine the Random Forest.

5. Explainable AI Methods

- Methods like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) could further clarify how the model arrives at its predictions, fostering trust and transparency in practical deployments.
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